

Safe Routes to School (SRTS)

Executive Summary

Client: Harriet Jackson
Student Consulting Team

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Background

The Safe Routes to School (SRTS) program is housed within Pittsburgh's Department of Mobility and Infrastructure (DOMI), serving Pittsburgh Public Schools. Run solely by Program Coordinator Harriet Jackson, SRTS increases student safety through education, encouragement, and infrastructure improvements for walking and biking routes. The program operated with minimal technology infrastructure with no formal system for managing audit data or generating stakeholder reports.

Project Description

Project Opportunity

SRTS faced three interconnected inefficiencies in its walkability audit process: duplicate data entry from paper to digital consumed significant staff time; manual report compilation delayed advocacy when stakeholders needed evidence for funding decisions; and historical audit data scattered across Google Docs, email threads, and physical storage prevented strategic planning and program-wide analysis.

Project Vision

We addressed these inefficiencies by building a two-part digital system on existing infrastructure: ArcGIS Survey123 for structured digital data collection, and a deployed web application using a large language model to automatically generate stakeholder-ready reports. The solution was designed to free Harriet from duplicative administrative work, enabling her to expand the program's reach and present SRTS as a data-driven safety program.

Project Outcomes

People: Harriet can now run a complete audit data workflow—from survey collection through report generation—without manual transcription or IT involvement. Seven audit participants at the March 26 Pittsburgh Morrow pilot completed the digital workflow with no prior training and no paper forms collected.

Process: Data collection was restructured into a pre-audit form (completed by school staff before the walk) and a post-audit form (completed by all participants afterward), a change driven directly by pilot feedback showing that filling out forms while walking, observing, and conversing was impractical. This restored the audit's core function as a collaborative site visit while still capturing structured data from all roles.

Technology: Two live Survey123 forms are deployed in Harriet's existing DOMI ArcGIS account, with role-specific questions branching across five participant types. A deployed web application pulls audit responses via the ArcGIS REST API, passes them to an LLM, and generates three audience-specific Word documents (DOMI Internal, School & Community Partner, and General Public) in under five minutes. Reports download as editable .docx files, preserving Harriet's editorial control before distribution.

Project Deliverables

Deliverables include two live ArcGIS Survey123 forms (pre- and post-survey) hosted in Harriet's DOMI account, the `srts_schools` feature layer serving as controlled vocabulary for both surveys, a deployed report generation web application, a GitHub repository with full source code, three sample reports generated from real Morrow pilot data and reviewed by Harriet, a Client Handoff Guide covering all workflows for independent operation, and a Technical Manual for DOMI IT covering system architecture, API dependencies, and credential management.

Recommendations

Three extensions would meaningfully increase the system's long-term value. An ArcGIS Dashboard connected to existing feature layers would give Harriet a live view of program activity across audits without requiring a new report for each meeting. A year-end summary report, added as an additional option in the existing pipeline, would allow Harriet to communicate program-wide impact to decision-makers who control the budget. Finally, automating post-survey distribution and submission reminders can help Harriet with improving submission rates and report completeness.

Student Consulting Team

Evelyn Lui served as Client Relationship Manager and Project Manager, leading all client communications and designing both Survey123 forms.

Lillian Zhao led the technical implementation of the report generation deliverable, building and deploying the web application end to end.

Doris Gao led documentation and system design across both deliverables and participated in the March 26 pilot audit.